

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Gravity and Magnetic Methods of Prospecting
L-T-P: 3-1-0 Credits:4

3. Pre-requisites: NA

4. Syllabus and reference books:

Syllabus:

Principles of gravity, Gravitational Potential, Gravity of the Earth, Gravity method, nature of gravity and its variation, measurement of absolute and relative gravity, stable and unstable gravimeters, various corrections applied to gravity data, field surveys and reduction of data, Gravity Data Processing, Free air, Bouguer and Isostatic anomalies, gravity effects of simple geometrical shapes, Gravity anomalies due to 1-D, 2-D and 3-D bodies.

Principles and Elementary Theory of Magnetic method, Nature of earth's magnetic field and its variation, magnetic field balance, Field Instruments for Magnetic Measurements, field surveys and reduction of magnetic data, magnetic effects of simple geometric shapes, magnetic anomalies due to 1-D, 2-D and 3-D bodies

Qualitative and quantitative rules for the gravity and magnetic data interpretation, problems of ambiguities and resolution in the interpretation, regional and residual anomalies and its separation using various methods, their relative merits and demerits, second vertical derivative, various methods of its determination, their merits and demerits, determination of anomalous mass from residual gravity, excess mass, determination of upward and downward continuation and their uses, Poisson's relation, pseudo-gravity and pseudo-magnetic transformations, role of remanent magnetisation, Data enhancement Techniques, Reduction to pole for magnetic data, indirect methods of interpretation using characteristic curves and polygon method of interpretation, limiting depth rules, aero-gravity and aeromagnetic surveys, borehole gravity method, micro-gravity surveys.

Space for entering the syllabus

Reference Books:

1. Applied Geophysics- Telford, Geldart and Sherifi
2. Introduction Geophysical Prospecting- Dobrin, Savit
3. Gravity and Magnetic Exploration Principles Practices and
4. applications- Hinze, Ralph and Saad
5. Applied Geophysics- Burger, Sheehan and Jones
6. Fundamentals of Geophysics- Lowrie
7. The Solid Earth by C.M.R. Fowler
8. Looking into the Earth by Alan E . Musset and M. Aftab Khan

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5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Principles of gravity, Gravitational Potential, Gravity of the Earth, Gravity method, nature of gravity and its variation	4
2.	Measurement of absolute and relative gravity, stable and unstable gravimeters	4
3.	Various corrections applied to gravity data, field surveys and reduction of data,	4
4.	Gravity Data Processing, Free air, Bouguer and Isostatic anomalies, gravity effects of simple geometrical shapes, Gravity anomalies due to 1-D, 2-D and 3-D bodies.	6
5.	Principles and Elementary Theory of Magnetic method, Nature of earth's magnetic field and its variation, magnetic field balance	4
6.	Field Instruments for Magnetic Measurements, field surveys and reduction of magnetic data, magnetic effects of simple geometric shapes, magnetic anomalies due to 1-D, 2-D and 3-D bodies	6
7.	Qualitative and quantitative rules for the gravity and magnetic data interpretation, problems of ambiguities and resolution in the interpretation, regional and residual anomalies and its separation using various methods, their relative merits and demerits, second vertical derivative, various methods of its determination, their merits and demerits, determination of anomalous mass from residual gravity, excess mass	8
8.	Determination of upward and downward continuation and their uses, Poisson's relation, pseudo-gravity and pseudo-magnetic transformations, role of remanent magnetisation	4

9.	Data enhancement Techniques, Reduction to pole for magnetic data, indirect methods of interpretation using characteristic curves and polygon method of interpretation	4
10.	Limiting depth rules, aero-gravity and aeromagnetic surveys, borehole gravity method, micro-gravity surveys.	4
Total number of hours		48